

TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES – MANILA

Smart Trash Bin Sensor:

A Prototype for Cleaner Communities

Science, Technology, and Society
GEC 007 / CPE21S2

SUBMITTED BY:
Gregorio, Fausto Kenneth
Sardin, Vale Daniel
Elento, Joel
Basinillo, Jewel Heart

SUBMITTED TO:
Dr. Lagumbay, Samuel Jr.

October 29, 2026

Table of Contents

I. Introduction..... 3

II. Objective of the Study.....3

III. Methodology.....4

IV. Data Gathering and Results..... 4

V. Proposed Prototype..... 4

VI. Significance of the Study.....9

VII. Scope and Limitations.....10

VIII. Conclusion..... 10

IX. Recommendation..... 11

X. References.....12

XI. Documentation..... 12

I. Introduction

Overflowing trash bins are a common problem in households, schools, and public areas, particularly in communities like Barangay Santa Ana, Pateros. When bins are not emptied on time, garbage spills onto the streets, attracts pests, causes bad odors, and can clog drainage, which adds to health and flooding risks. While the smart trash bin sensor prototype serves as the technological solution, this research is driven by the local waste management needs of Barangay Santa Ana, Pateros. Through science and technology, a trash bin sensor can help show when a bin is almost full so it can be emptied before it overflows, while a hands-free lid makes the bin cleaner to use since people no longer have to touch it.

II. Objective of the Study

The main objective of this study is to develop a sensor-based trash bin prototype that can help improve waste management in households and other areas where trash bins are used. Specifically, it aims to:

1. Determine the fill level of a trash bin using an ultrasonic sensor to detect when the bin is nearly full.
2. Develop an alert system using an Arduino Uno and a buzzer to warn users when the trash reaches a predetermined fill level.
3. Develop an automatic lid mechanism using a second ultrasonic sensor and a servo motor to open and close the lid when a hand or object is detected, provided that the bin is not full.

4. Evaluate the functionality of the prototype in detecting the trash level, activating the alarm, and controlling the automatic lid mechanism;
5. Demonstrate the potential contribution of the prototype to preventing overflowing trash bins and promoting cleanliness and public health; and
6. To gather and analyze waste management data from Barangay Santa Ana, Pateros, to assess the feasibility of the prototype in a real-world community setting.

III. Methodology

To ensure the prototype addresses real-world concerns, the research includes a data-gathering process involving key informants from Barangay Santa Ana, Pateros. The study was employ through a qualitative approach, utilizing structured interviews with Barangay Tanods and local community members to gain insights into current waste management practices, specific areas prone to overflowing trash, and community requirements for a more sanitary disposal system. The insights gathered from these interactions will serve as the basis for the prototype's design requirements and operational features, ensuring that the technology is aligned with the actual needs of the community.

IV. Data Gathering and Results

The interview revealed that the barangay already has an established waste management system, with garbage collection conducted early in the morning and personnel monitoring the

collection process. According to the interviewee, the area generally becomes clean after the garbage collection truck passes. The interviewee also mentioned that CCTV cameras are used to monitor garbage-related conditions.

Despite the existing system, some waste management challenges remain, particularly in collecting oversized and heavy materials such as beds, wood, and stones. These materials are not always collected during regular garbage collection and may require separate arrangements or designated collection schedules. As a result, some waste may accumulate while waiting for collection. The interviewee indicated that ordinary garbage collection is generally manageable in their area, partly because of the presence of monitoring personnel.

Regarding the proposed smart trash bin system, the interview explored the possibility of sending notifications when bins become full or when special waste requires attention. However, the interviewee's response did not clearly establish whether the proposed notification system would be beneficial. The discussion also identified the possibility of placing trash bins in areas where garbage frequently accumulates, with several locations mentioned as potential sites. Overall, the interview provided information about the barangay's existing waste management practices, remaining collection challenges, and possible considerations for implementing the proposed smart trash bin system.

V. Proposed Prototype

The proposed prototype is a flip-top trash bin with two ultrasonic sensors, an Arduino Uno microcontroller, a servo motor, a buzzer, and a power supply. The first sensor (Sensor A) is placed on the front of the bin and detects a hand or object approaching the lid. The second sensor (Sensor B) is mounted under the lid, facing down, and measures the distance to the trash inside.

The servo motor is attached at the lid hinge and moves the lid. The Arduino Uno reads both sensors and decides what to do:

1. Sensor B measures the distance from the lid to the trash. If the distance is at or below the set threshold (about 10 cm), the bin is considered full. The buzzer sounds, and the lid stays closed.
2. If the bin is not full, the Arduino checks Sensor A. When it detects something within about 20 cm, the servo swings the lid open.
3. If the lid stays open for a few seconds, then the servo closes it again.
4. If once the lid is closed, Sensor B takes a new reading and the cycle repeats.

Component	**Quantity**	**Purpose**
Arduino Uno R3 with USB Cable	1	Main Controller
HC-SR04 Ultrasonic Sensor	2	Hand Detection (A) and Fill Level Detection (B)
SG90 micro servo (180°)	1	Opens and closes the lid
Active Buzzer	1	Full-bin Alarm
Breadboard and Jumper Wires	1	Connections
Flip-top trash bin	1 set	Prototype Body
5V Power Supply (Power Bank)	1	Power Source

Table 1. Components of the Smart Trash Bin Sensor Prototype

The table above lists the necessary hardware components required to construct the smart trash bin prototype.

Component	**Pin**	**Arduino Uno**
Sensor A (hand)	VCC/GND	5V/GND
Sensor B (hand)	TRIG/ECHO	D2/D3
Sensor A (fill level)	VCC/GND	5V/GND
Sensor B (fill level)	TRIG/ECHO	D4/D5
Servo SG90	Red/Brown	5V/GND
Servo SG90	Orange (Signal)	D9
Buzzer	+/-	D8/GND

Table 2. Pin Connections of the Components to the Arduino Uno

Table 2 outlines the specific wiring connections between the sensors/actuators and the Arduino Uno microcontroller for proper system operation.

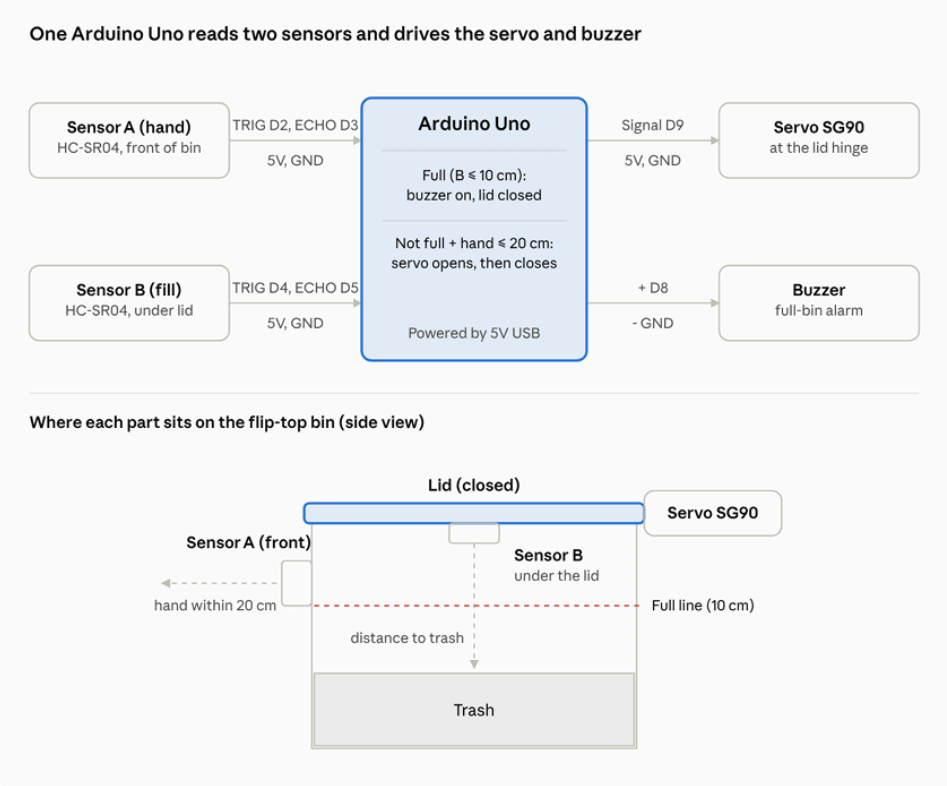


Fig.1. Block and Wiring Diagram of the Smart Trash Bin Sensor

In Figure 1. Sensor B, placed under the lid, checks the fill level first. It measures the distance from the lid to the trash. If the trash exceeds the distance of 10 cm or less, it automatically detects that the lid is already full. After the detection, the Arduino turns on the buzzer and keeps the lid closed.

If the bin is not yet full, Sensor A, at the front of the bin, checks for a hand. When a hand enters the range of Sensor A at 20 cm, the Arduino signals the servo motor to open the lid itself. After a few seconds, the servo motor closes the lid again.

Reference Designs

The photos below are reference designs used as a blueprint for the lid, servo, and front sensor layout. They are not our finished prototype.

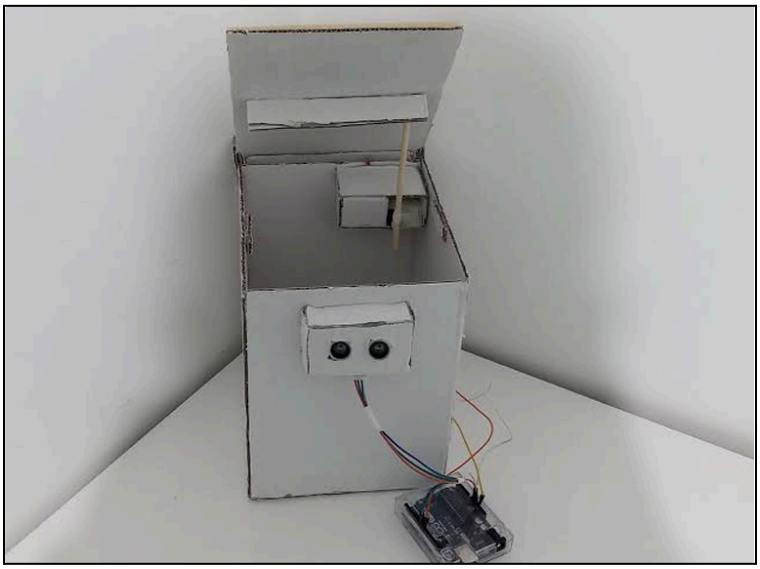


Fig. 2. Cardboard Bin with a Servo-Driven Lid and a Front-Mounted Ultrasonic Sensor



Fig. 3. The Front Sensor Detects a Hand or Object so the Lid Can Open

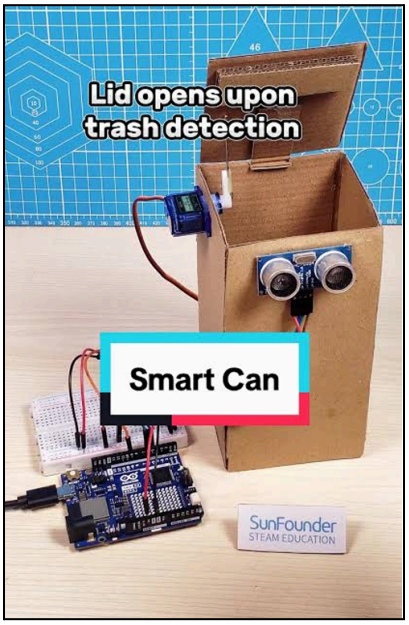


Fig. 4. SunFounder "Smart Can", Where the Lid Opens upon the Trash Detection

VI. Significance of the Study

The project Smart Trash Bin Sensor demonstrates how technology can support timely garbage collection. Specifically, it aims to help the following:

- **Households**. The hands-free lid reduces contact with the bin, which helps keep hands clean.
- **Students**. It promotes awareness of proper waste disposal.
- **Schools**. It promotes awareness of proper waste disposal and the importance of keeping the community clean and healthy.
- **Barangays**. It demonstrates how technology can support timely garbage collection.

VII. Scope and Limitations

The project focuses on detecting the fill level of a trash bin, opening the lid automatically when the bin is not full, and activating a buzzer when a predetermined threshold is reached. Its performance depends on sensor quality, calibration, placement, and power supply. Uneven piles of trash or soft items may affect the readings. The SG90 servo is light-duty, so it is only suited for a small and lightweight lid. The fill level is only measured while the lid is closed. The prototype does not compact the trash, notify collectors remotely, or empty the bin automatically. It should not replace regular garbage collection or established waste management procedures.

VIII. Conclusion

The Smart Trash Bin Sensor demonstrates the application of science and technology in improving waste management through fill-level detection, audible alerts, and an automatic lid mechanism. However, its functionality depends on proper sensor calibration, placement, and a reliable power supply. The prototype may also experience inaccurate readings due to uneven trash surfaces or soft materials, while its SG90 servo limits the weight of the lid it can operate. Since the fill level is measured only when the lid is closed, continuous monitoring is not possible. The prototype cannot compact trash, remotely notify collectors, or empty the bin automatically. Therefore, the system serves as a supplementary tool for waste management and should not replace regular garbage collection and established waste disposal practices.

IX. Recommendation

This study reveals the potential of the Smart Trash Bin Sensor as a technology-based solution for improving waste management and promoting cleanliness in the community. Thus, the following recommendations are hereby presented:

- Users should ensure that the prototype has a reliable power supply, regularly check its functionality, and empty the bin when the buzzer indicates that it is nearly full.
- Schools and barangays may consider using the prototype to promote cleanliness, proper waste disposal, and awareness of technology-based waste management solutions.
- Waste collectors should maintain regular garbage collection schedules and follow proper waste disposal procedures, as the prototype cannot automatically empty the bin or send remote notifications.
- Future researchers should improve sensor accuracy through proper calibration and placement, explore stronger servo motors and reliable power sources, and consider adding continuous monitoring, wireless notifications, odor control, and waste segregation to enhance the prototype's functionality.
- The prototype should undergo an evaluation for testing the accuracy of the sensor, durability of the prototype, reliability of the automatic lid, and the effectiveness of its alert system.

X. References

Arduino Making. (2022, May 27). *How to make an automatic trash bin using cardboard*. Arduino Project Hub. https://projecthub.arduino.cc/arduinomaking/how-to-make-an-automatic-trash-bin-using-cardboard-7c4a8d

Ashraf, M. (2018, May 29). *Arduino trash bot - Auto open/close trash bin*. Hackster.io. https://www.hackster.io/ashraf-minhaj/arduino-trash-bot-auto-open-close-trash-bin-5b1c92

SunFounder. (2023, June 29). *Smart Can: Lid opens upon trash detection* [Video]. YouTube. https://www.youtube.com/shorts/JbTtRU4-BpU

XI. Documentation

Assigned Task	Assigned Member	Date	Image	Remarks

XII. INTERVIEW TRANSCRIPTION

(00:00 - 1:00mins)
[interviewer] :So Ano ano po mga karaniwang problema sa basura sa inyong barangay po at pano po ito naaapektohan ang mga residente

[person1] :Dito naman sa ano.. Dito sa pinaka barangay kase meron naman nag lilinis dito. Uh.. six O clock ng umaga lalabas na sila umh and mga basura sa loob hinihilera sa labas nila,..... para meron sila kasi… nilalabas nalang nila. Maaga nilalabas ang basura dito, lahat ng basura dito sa Santa…., and pinaka main namin na problem ay yung nasa looba

:uhmm….

(1:01 -2:01mins)
[person1] :pero ang ginagawa naman namin ay nilalabas nila at kinukuha. Dito samin sa lugar ko meron tagabantay. Merong bantay …….. Nag babantay sila ng alas kwatro ng madaling araw minsan palang alas tres palang andyan na sila, binabantayan nila yung.. nag cocollect ng basura kaya pagkaalis ng basura…. Pag alis ng truck malinas ulit ang lahat ganun ang ginagawa dito samin..

[intgerviewr] :Next po is pano niyo po minomonitor mga basurahan sa barangay po… katulad po sabi nyo pano nyopo nalalaman kung puno na ang mga to

[person1] :Nakikita naman sa CCTV

[interviewr] :Uh okay thank you po…then next question po, ano ano po ang mga problema naranasan ninyo sa pagkuha ng mga basurahang ito at tiyak na nakokolecta ng tama ang basurang ito.

(2:02-3:29mins)
[peson1] :Kunti nlang problema namin dto po yung pag nag labas sila ng malaking waste uhmhm… yung ang problema malaking waste. Halimbawa mag lalabas sila ng kama eh hindi nmn basta basta kukunin yun may ano yun …. May kanya kanya kasi yun.. kanyang araw nga na kukunin yun ang problema.. Yun nga. Halimbawa dito samin sa friday kukunin yung mga iba nag tatapon palang mga marter miyerkules so naiimbak ang basura….. Halimbawa nag ano sila uhm nag aano ng mga ano kahoy, nilalabas nila yun hindi kinukuha yun tsaka yung bato hindi kinukuha yun ng mangbabasura… kahit hintayin nila pag dating nung dump waste na truck hindi kinukuha ang mga yun.. Yun ang problema ko.. hindi masyado maproblema dito ang basura sa ngayon kasi yung ating

mga bantay nayun, siguro almost twenty ata yun ih mga madaling araw nadyan na… malayo namn yung nararating nung ating mga ……

(3:30-4:23mins)
[interviewer] :Next question po… sa ginagawa po naming project sa tingin po ninyo makakatulong po ba ang makatanggap ng text o notification kapag puno na ang mga basurahan or… mga basura po na sobrang bigat

[person1]:hmm… hindi naman …….. Paonline yang sa inyo

[interviewer] :Ano po yung basurahan po is inonotif po kayo na if puno na or may mabigat na object na sobrang laki na

[person1]:inaaano parin……. Kami ay inaano sakanila……kung kami ay di mapapalitan… problem din yan kung hindi mapalitan kasi halimbawa minsan punong puno na yung basura basura hindi mapalitan mabutas pa, kaya lang hindi nila na ikakalat kasi meron na kami kanya kanya, kay ilagay nyo lang yan…..

(4:24-5:29:mins)
[interviewer] :Saan pong lugar sa inyong barangay ay satingin nyo ay magiging pakipakinabang ang trashbin po namin na mag nonotif po sa inyo kung puno napo sya or may mabigay po na gamit…

[person1] :mag lalagay kayo ng trash bin….

[interviewer] :I mean if ever na dito po namin yun … gano po kalaki yung tulong na manonotif kayo

[person1] :siguro sa "*esperanza"* diba, "*glenn"*

[person2] :Poo…

[person1] :Transbin daw nila san ilalagay pwede sa *"esperanza"* diba

[person2] :Pwede naman po

[person1] :*"esperanza"*... sabagay don sa "*esperanza"* dalawa tayong nag aano eh… *"san roqoue"* tsaka *"santa ana"*

[person2] :Ilan ba ang ilalagay mo

[person1] :Ilan ba ang ilalagay mo……

[interviewer] :Ano lang po sya research palang po sya.. prototype …

[person1] :Ah okay okay…

[person1] :Pag nandyan nayan marami tayong pag lalagyan…. Kung saan natin ipwepwesto, doon naitn ipwepwesto sa laging maraming basura

[interviewer] :Yun lang po thank you po

[person1] :Sige sige salamat

[interviewer] :Thank you po..,